30. Ans: D

For the cut-off wavelength,

$$eV = \frac{hc}{\lambda_{min}}$$

Since the λ_{min} for graph 2 is halved of graph 1, that means that the accelerating potential for graph 2 is doubled that of graph 1.

Since the characteristic wavelength remain the same for graph 1 and graph 2, this means that the target metal is the same.